

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=6 njev=6

message='gtol' termination condition is satisfied.

reported optimality=8.306048471240501e-07

initial={'cost': 583.2574682595425, 'gradient_inf_norm': 0.06132937771576508, 'gradient_l2_norm': 0.09525936213437144, 'jacobian': {'condition_number_nonzero_subspace': 323.4036682465926, 'numerical_rank': 14, 'numerical_tolerance': 1.669787750700851e-10, 'singular_values': [191.54496922680008, 174.73261076399075, 152.66820856598744, 48.523762813532024, 44.75982314756852, 25.941212101153077, 13.174640035842126, 11.996761150742078, 11.66429702660792, 10.772021077649892, 9.272600435443676, 3.8170572217885956, 3.7876329497550585, 0.5922782826345329]}, 'residual_l2_norm': 34.15428137904654}

final={'cost': 583.2574676144673, 'gradient_inf_norm': 8.306048471240501e-07, 'gradient_l2_norm': 1.0965989108637812e-06, 'jacobian': {'condition_number_nonzero_subspace': 324.10067165340627, 'numerical_rank': 14, 'numerical_tolerance': 1.6698028029742595e-10, 'singular_values': [191.54669590568224, 174.73297665244993, 152.66856326722535, 48.523895868153424, 44.76010117453063, 25.941367720166127, 13.174576775182857, 11.996850138089409, 11.664639101686367, 10.77224380567949, 9.272816986174188, 3.8170937432826393, 3.787617803027176, 0.5910098702619245]}, 'residual_l2_norm': 34.15428136015944}

coordinate_step_l2_norm=0.0010755743343700917

cost_reduction=6.450752607634058e-07

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=4.710650673165391e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 181.28757272400682, 'difference_l2_norm': 8.539824264890797e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 181.28757272405292, 'relative_error': 4.710650673165391e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.